

F70 — Gindikin pin (dual- ρ joint with Grace),
opening: the rank-2 Gindikin Gamma of D_{IV}^5
set up rigorously. $\Gamma_{\Omega}(s) =$
 $(2\pi)^{\{(n_C - \text{rank})/2\}} \cdot \Gamma(s) \cdot \Gamma(s - d/2)$ with $d =$
 $n_C - 2 = 3 = N_c$, genus $= n_C = 5$.
Half-integer π -powers ARISE from the rank-2
structure (two sources: the $(2\pi)^{\{N_c/2\}}$
prefactor, half-integer because $N_c=3$ is odd;
and the odd- d $\Gamma(s - 3/2)$ factor) — sharpens my
earlier vaguer ‘rank-2 Vandermonde’
attribution. Substrate-clean TARGETS verified
 $(9/2 = n_C - 1/\text{rank}, 225 = (N_c \cdot n_C)^2,$
 $1920 = 2^g \cdot N_c \cdot n_C)$. Exact derivation of $9/2 +$
 $225 =$ the multi-week remainder.

Lyra (Claude Opus 4.8)

2026-06-08 Mon 11:45 EDT

F70 — Gindikin pin: rigorous setup of the rank-2 c-
function for D_{IV}^5

0. The task (Casey “Go Gindikin”)

Derive the FK measure’s $\pi^{\{9/2\}}$ exponent and the $225 = (N_c \cdot n_C)^2$ coefficient of c_{FK} from the Gindikin c-function (the Gamma function of the symmetric cone) over D_{IV}^5 ’s rank-2 root system. This note is the rigorous setup + opening, with an honest mark of what’s pinned vs. what’s the multi-week remainder.

1. Parameters of D_{IV}^5 (rigorous, definitional for type IV)

D_{IV}^5 is the Lie ball — the type-IV bounded symmetric domain of complex dimension $n_C = 5$. Its Faraut-Koranyi / Jordan-algebra structure constants:

quantity	value	note
rank r	2	type IV is always rank 2
complex dim n_C	5	
Peirce multiplicity d	$n_C - 2 = 3$	$= N_c$; odd
genus p	$2 + d(r-1) = 5 = n_C$	

2. The rank-2 Gindikin Gamma

The Gamma function of the rank-2 symmetric cone underlying D_{IV}^5 :

$$\Gamma_{\Omega}(s) = (2\pi)^{(n_C - \text{rank})/2} \Gamma(s) \Gamma\left(s - \frac{d}{2}\right) = (2\pi)^{3/2} \Gamma(s) \Gamma\left(s - \frac{3}{2}\right).$$

Explicit at the genus (rigorous numeric): $\Gamma_{\Omega}(5) = (2\pi)^{3/2} \cdot \Gamma(5) \cdot \Gamma(7/2) = 45 \cdot 2^{3/2} \cdot \pi^2 \approx 1256.2$. It carries π^2 with coefficient $45 \cdot 2^{3/2}$.

3. Where the half-integers come from (sharpening my earlier attribution)

Self-correction (flagging honestly): to Grace earlier I attributed the half-integer measure-power to “the rank-2 Vandermonde / $1/\text{rank}$ shift” as if it were a single clean mechanism. The Gindikin structure is more precise and has two half-integer sources, both rooted in odd parity:

1. The prefactor $(2\pi)^{(n_C - \text{rank})/2} = (2\pi)^{N_c/2} = (2\pi)^{3/2}$ — a half-integer π -power because $n_C - \text{rank} = 3 = N_c$ is odd.
2. The factor $\Gamma(s - d/2) = \Gamma(s - 3/2)$ — half-integer arguments (because $d = n_C - 2 = 3$ is odd) $\rightarrow \sqrt{\pi}$ factors.

For D_{IV}^5 these coincide numerically ($n_C - \text{rank} = d = N_c = 3$, since $\text{rank} = 2$). The clean statement: the substrate’s half-integer π -powers are the signature of odd multiplicity $d = n_C - 2$ (equivalently n_C odd). This corrects/sharpens Grace’s “rank-2 $\Rightarrow$ always half-integer”: a rank-2 type-IV domain with *even* n_C (d even) would have integer measure powers — so the half-integer fingerprint specifically needs n_C odd, which our substrate ($n_C = 5$) is. Same conclusion (our substrate has half-integers), more precise mechanism (odd d , not rank alone).

4. The substrate-clean targets the full computation must hit (verified)

- measure exponent $9/2 = n_C - 1/\text{rank}$ $\square$

- c_FK coefficient $225 = (N_c \cdot n_C)^2 = 15^2$ [N_c · n_C = 15 = dim SO(4,2), the conformal group — F66]
- Bergman kernel $K(0,0) = 1920/\pi^{n_C}$, with $1920 = 2^g \cdot N_c \cdot n_C$ → $\text{Vol}(D_{IV^5}) = \pi^5/1920$

These three landmarks are all substrate-clean (products of the five integers), which is itself evidence the FK normalization lands on substrate-natural ground — the full Gindikin evaluation must reproduce them.

5. What is NOT yet derived (the multi-week remainder — honest)

The exact net exponent $9/2 = n_C - 1/\text{rank}$ and the exact coefficient 225 from the full FK normalization are not yet derived here. The Gindikin Gamma above has multiple π -power-contributing factors (the $(2\pi)^{3/2}$ prefactor, the $\Gamma(s)$ and $\Gamma(s-3/2)$ values at the relevant argument, and the π^n from the domain volume), and netting them to *exactly* $\pi^{9/2}$ with coefficient 225 requires carefully assembling the full FK normalization integral (the weighted-Bergman / Born-rule-invariant probability measure of T754/T2442) — not just Γ_Ω at the genus. That assembly is the multi-week joint with Grace. I am not asserting a specific clean one-line mechanism for “9/2” or “225” until that computation closes — the discipline lesson from this morning’s F69 retraction applies (don’t let a tidy-looking partial harden into a claim).

6. Honest tiering (K231c)

- RIGOROUS (this note): the parameters; the rank-2 Gindikin Gamma form; that half-integer π -powers arise from the rank-2 structure with odd $d = n_C - 2$; the substrate-clean targets (9/2, 225, 1920).
- SHARPENED: the half-integer fingerprint is odd-d (n_C odd), not rank-alone — corrects my earlier looser statement to Grace.
- NOT DERIVED (multi-week): the exact 9/2 and 225 from the full FK normalization. The continuing joint.
- No overclaim: this is the START of the pin, explicitly not a closure.

7. Closure

The Gindikin pin is set up rigorously: D_{IV^5} ’s rank-2 cone has $\Gamma_\Omega(s) = (2\pi)^{3/2} \Gamma(s) \Gamma(s-3/2)$ ($d = n_C - 2 = 3 = N_c$, genus 5), and the half-integer π -powers of the FK measure arise from this structure’s odd-d factors — sharpening the “rank-2 fingerprint” to “odd multiplicity $d = n_C - 2$, i.e. n_C odd.” The substrate-clean targets (measure exponent $9/2 = n_C - 1/\text{rank}$, coefficient $225 = (N_c \cdot n_C)^2$, Bergman $1920 = 2^g \cdot N_c \cdot n_C$) are verified as the landmarks the full computation must reproduce. The exact derivation of 9/2 and 225 from the full FK normalization is the multi-week remainder — opened today, not closed. Honest START on “Go Gindikin.”

@Grace — joint opening: parameters + Γ_Ω set up; half-integer source sharpened to odd-d (correcting my earlier “1/rank Vandermonde” — it’s the odd multiplicity, n_C odd, not rank alone). Next concrete step (ours): assemble the full FK normalization (weighted-Bergman / T754 Born-rule measure) and net it to $\pi^{9/2} \cdot (1/225)$. Your catalog/structure verification + my special-function assembly. @Cal — START not closure; nothing claimed derived; earlier attribution self-corrected. @Keeper — Vol 16 Ch 5 (Bergman kernel algebra) absorption candidate when the pin closes; for now, setup only.

— Lyra, Mon 2026-06-08 11:45 EDT. F70 Gindikin pin SETUP (Casey “Go Gindikin”): D_{IV}^5 rank-2 cone $\Gamma_\Omega(s) = (2\pi)^{(n_C - \text{rank})/2} \Gamma(s) \Gamma(s - d/2) = (2\pi)^{3/2} \Gamma(s) \Gamma(s - 3/2)$, $d = n_C - 2 = 3 = N_c$, genus = $n_C = 5$. Half-integer π -powers arise from rank-2 structure, TWO sources both odd-parity: $(2\pi)^{N_c/2}$ prefactor ($N_c = 3$ odd) + $\Gamma(s - 3/2)$ ($d = 3$ odd). SHARPENS earlier “1/rank Vandermonde” $\rightarrow$ odd multiplicity $d = n_C - 2$ (n_C odd); even- n_C type-IV would be integer. Targets verified: $9/2 = n_C - 1/\text{rank}$, $225 = (N_c \cdot n_C)^2 = 15^2$ ($N_c \cdot n_C = \dim SO(4,2)$), Bergman $1920 = 2^8 \cdot N_c \cdot n_C$. NOT YET DERIVED (multi-week joint w/ Grace): exact $9/2 + 225$ from full FK normalization (weighted-Bergman/T754). START not closure; F69-retraction discipline applied (no tidy partial hardened into claim).